

What is Chunking

The chunking is the process of splitting text/Documents into smaller lines is called chunking.

When to use chunking:-

You can use chunking in RAG ^{based} application when

- * Document / PDFs are too large to LLMs
- * To improve retrieval accuracy.
- * Reduce Token cost and improve performance.

There are Techniques to chunk the text.

(i) Fixed Size Chunking:-

⇒ Split text into Fixed Size length of character or tokens.

pros ⇒ Very simple and fast to implement with consistent size.

cons ⇒ It can split the meaning (or) break the particular meaning, which will affect the retrieval accuracy.

(ii) Recursive chunking:

separator = ["\n", " ", ",", "'"]

It is a hierarchical text-splitting strategy for RAG. It breaks down the text ~~into~~ by separators. Once it reaches the chunk size then it will stop splitting. This balances both ~~relevant~~ text split and chunk size.

Pros: Balance both & context very well.
most popular approach.

Cons:

little complex than the Fixed size chunking and slower during processing.

(iii) Sentence chunking:

Split text document along natural sentence boundaries rather than token counts.

This preserve the grammatical structure and coherent thoughts. it gives complete meaning to par rather than broken sentence.

Pros: Keep sentence intact and is easy to understand.

Cons: Chunk size vary, and related info may end up with different chunks.

(iv) Semantic chunking

Semantic chunking splits documents at natural topic shifts using embedding similarity rather than fixed token length.

This preserves context and prevents ideas rather than being cut in half. It increases the retrieval precision.

Canchain offers built-in semantic splitter to automate the process.

pros:- produce highly meaningful chunks and improve retrieval quality.

cons:- More computationally expensive and slower than rule-based methods.